

Natural language statement

Domain

GroundTruth

Exercise 10.4.2. Let $\Omega = \mathbb{R}^d$ or $\Omega = \mathbb{R}^d_+$ and assume that (2) holds for any u in $C_0^\infty(\Omega)$ and some k, m, p, q with a constant independent of u. Then prove that $k \geq m$ and (1) holds. Notation: The displays referred to are those of Lemma 10.4.1, where k in $\{1, 2, \dots\}$, p in $[1, \infty]$, m in $\{0, \dots, k\}$ and q in $(0, \infty]$: $k - (d)/(p) = m - (d)/(q)$ $[u]_{W_q^m(\Omega)} \leq N [u]_{W_p^k(\Omega)}$ Here $[u]_{W_p^k(\Omega)} = \sum_{ \alpha \leq k} \ D^\alpha u\ _{L_p(\Omega)}$ $[u]_{L_p(\Omega)}$ is the top-order seminorm - the sum runs over multi-indices of order exactly k - and $\mathbb{R}^d_+ = \{x \in \mathbb{R}^d : x^1 > 0\}$. In the exercise k, m, p, q are not assumed to satisfy any relation that they must be the point. For an integer $k \geq 0$, C_0^k denotes the C^k functions on $\mathbb{R}^d$ that vanish for x sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_i u$ and $u_{x^i x^j} = D_{ij} u$. Repeated indices are summed. $L_p = L_p(\mathbb{R}^d)$ is taken with respect to Lebesgue measure.	PDE	<pre> noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) -> F) : EuclideanSpace R (Fin d) -> F := fun x => fderiv R u x (EuclideanSpace.single i 1) noncomputable def multiDeriv (alpha : Fin d -> N) (u : EuclideanSpace R (Fin d) -> F) : EuclideanSpace R (Fin d) -> F := (List.finRange d).foldr (fun i v => (partialDeriv i)^[alpha i] v) u theorem krylov_sobolev_10_4_2_seminorm_embedding_forces_scaling_relation {d : N} [NeZero d] {Omega : Set (EuclideanSpace R (Fin d))} (hOmega : Omega = Set.univ \ / Omega = {x : EuclideanSpace R (Fin d) 0 < x 0}) {k m : N} (hk : 1 <= k) {p q : R} (hp : 1 <= p) (hq : 0 < q) (hemb : exists N : R >= 0, forall u : EuclideanSpace R (Fin d) -> R, ContDiff R inf u -> HasCompactSupport u -> tsupport u subset Omega -> (sum alpha in Finset.piAntidiag Finset.univ m, eLpNorm (multiDeriv alpha u) (ENNReal.ofReal q) (volume.restrict Omega)) <= N * sum alpha in Finset.piAntidiag Finset.univ k, eLpNorm (multiDeriv alpha u) (ENNReal.ofReal p) (volume.restrict Omega)) : m <= k /\ (k : R) - (d : R) / p = (m : R) - (d : R) / q := by sorry </pre>
Lemma 12.10.2. Let p in $(1, \infty]$ and let $0 < \delta := -d/p < 1$. Then there is a constant N such that, for any ϕ in S and x, y in $\mathbb{R}^d$, $\phi(x) \leq N (1 - \Delta)^{1/2} \phi (1 - \Delta)^{1/2} \phi (y) \leq N x - y ^\delta (1 - \Delta)^{1/2} \phi (1 - \Delta)^{1/2} \phi (y)$. Notation: S is the Schwartz space. By Definition 12.9.1, $(1 - \Delta)^{1/2}$ is the pseudo-differential operator with symbol $(1 + x ^2)^{1/2}$, that is $(1 - \Delta)^{1/2} \phi = F^{-1}((1 + x ^2)^{1/2} F \phi)$, where F is the Fourier transform. At $p = \infty$ the convention is $d/p = 0$, so $\delta = 0$.	PDE	<pre> noncomputable def besselop {d : N} (: R) : S(EuclideanSpace R (Fin d), C) -> L[C] S(EuclideanSpace R (Fin d), C) := SchwartzMap.fourierMultiplierCLM C fun xi => (((1 + (2 * Real.pi) ^ 2 * xi ^ 2) ^ (- d / 2) : R) : C) theorem krylov_sobolev_12_10_2_bessel_potential_holder_embedding {d : N} { R} {p : R >= 0inf} (hp : 1 < p) (hdelta0 : 0 < - d * (p^1).toReal) (hdelta1 : - d * (p^1).toReal < 1) : exists N : R, forall (phi : S(EuclideanSpace R (Fin d), C)) (x y : EuclideanSpace R (Fin d)), phi x <= N * (eLpNorm ((besselop phi) p volume).toReal /\ phi x - phi y <= N * x - y ^ (- d * (p^1).toReal) * (eLpNorm ((besselop phi) p volume).toReal := by sorry </pre>
Exercise 12.2.13. Prove that if the coefficients a^α of an mth order strongly elliptic differential operator are real and $d \geq 2$, then m is even. Notation: From Definition 12.2.1: let $m \geq 1$ be an integer and let a^α be some (complex) numbers given for any multi-indices α such that $\alpha \leq m$. The operator $L = \sum_{ \alpha \leq m} a^\alpha D^\alpha$ is called an mth order operator with constant coefficients. It is called (mth order) strongly elliptic if both $\sum_{ \alpha = m} a^\alpha x^\alpha \neq 0$ for x in $\mathbb{R}^d \setminus \{0\}$, $\sum_{ \alpha \leq m} a^\alpha x^\alpha = 0$ for x in $\mathbb{R}^d$. The polynomial $\sigma(x) = \sum_{ \alpha \leq m} a^\alpha x^\alpha$ is called the characteristic polynomial of L. Here $x^\alpha = (x^1)^{\alpha_1} \dots (x^d)^{\alpha_d}$.	PDE	<pre> def IsStronglyElliptic {d : N} (m : N) (P : MvPolynomial (Fin d) C) : Prop := P.totalDegree <= m /\ (forall xi : Fin d -> R, xi != 0 -> MvPolynomial.eval (fun j => ((xi j : R) : C)) (MvPolynomial.homogeneousComponent m P) != 0) /\ forall xi : Fin d -> R, MvPolynomial.eval (fun j => Complex.I * (xi j : C)) P != 0 theorem krylov_sobolev_12_2_13_real_strongly_elliptic_order_even {d m : N} (hd : 2 <= d) (hm : 1 <= m) (P : MvPolynomial (Fin d) C) (hp : IsStronglyElliptic m P) (hreal : forall alpha : Fin d -> 0 N, (P.coeff alpha).im = 0) : Even m := by sorry </pre>
Exercise 13.3.13. Prove that for any g in $H_p^{1/2}$ there exist $f_0, \dots, f_d$ in L_p such that $g = f_0 + \sum_j D_{j,j} f_j$ and $\sum_{j=0}^d f_j _{L_p} \leq N \ g\ _{H_p^{1/2}}$, where the constant N is independent of g. Also prove that if (7) holds with $f_0, \dots, f_d$ in L_p, then g in $H_p^{1/2}$ and $\ g\ _{H_p^{1/2}} \leq N \sum_{j=0}^d f_j _{L_p}$, where the constant N is independent of the f_j's. Notation: Throughout Section 13.3, p in $(1, \infty]$. By Definition 13.3.1, $H_p^1 = (1 - \Delta)^{1/2} L_p$ and, for g in H_p^1, $\ g\ _{H_p^1} = \ (1 - \Delta)^{1/2} g\ _{L_p}$. These are the spaces of Bessel potentials. In (7) the $D_{j,j}$ are distributional derivatives,	PDE	<pre> noncomputable def sobolevNorm {d : N} (s : R) (p : R >= 0inf) [Fact (1 <= p)] (g : S'(EuclideanSpace R (Fin d), C)) : R >= 0inf := f in {f : Lp C p (volume : Measure (EuclideanSpace R (Fin d))) bessePotential (EuclideanSpace R (Fin d)) C s g = f}, f e theorem krylov_sobolev_13_3_13_negative_order_divergence_decomposition {d : N} (p : R >= 0inf) [Fact (1 <= p)] (hp1 : 1 < p) (hp2 : p != top) : exists N : R >= 0, (forall g : S'(EuclideanSpace R (Fin d), C), MemSobolev (-1) p g -> exists f : Fin (d + 1) -> Lp C p (volume : Measure </pre>

Natural language statement

Domain

GroundTruth

so the identity is an identity of distributions, not of functions.

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(EuclideanSpace R (Fin d))),
  (g = (f 0 : S'(EuclideanSpace R (Fin d), C)) +
    (sum j : Fin d, d_{EuclideanSpace.single j (1 : R)}
      (f j.succ : S'(EuclideanSpace R (Fin d), C)))) /\
    (sum j, ||f j||e) <= N * sobolevNorm (-1) p g) /\
  forall (g : S'(EuclideanSpace R (Fin d), C))
    (f : Fin (d + 1) -> Lp C p (volume : Measure (EuclideanSpace R
      (Fin d)))),
    g = (f 0 : S'(EuclideanSpace R (Fin d), C)) +
      (sum j : Fin d, d_{EuclideanSpace.single j (1 : R)}
        (f j.succ : S'(EuclideanSpace R (Fin d), C))) ->
        MemSobolev (-1) p g /\ sobolevNorm (-1) p g <= N * (sum j,
          ||f j||e) := by
sorry
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Exercise 13.3.16. Sometimes it is hard to recognize whether a function u is in H_p^α , for $\alpha < 0$. Prove that if u has support in B_ρ , where ρ in $(0, \infty)$, and $|u(x)| \leq N_0 |x|^{-\nu}$, $\nu < d$, $0 < (\nu +)p < d$, < 0 , then u in H_p^α and $\|u\|_{H_p^\alpha}$ is less than a constant depending only on d, p, ρ, ν, N_0 . Observe that generally such a u not in L_p , because one need not have $\nu p < d$, and one cannot use the trivial embedding L_p subset H_p^α . By using Corollary 11 generalize the result and prove that if n in $\{0, 1, \dots\}$, in R , $< n$, u has support in B_ρ , $|D^\alpha u(x)| \leq N_0 |x|^{-\nu}$, for all $|\alpha| \leq n$, $\nu < d$, and either $< n$ and $0 < (\nu + - n)p < d$, or $= n$ and $\nu p < d$, then u in H_p^α and $\|u\|_{H_p^\alpha}$ is estimated by a constant depending only on d, p, ρ, ν, N_0 . Notation: Throughout Section 13.3, p in $(1, \infty)$, and H_p^α with its norm is Definition 13.3.1: $H_p^\alpha = (1 - \Delta)^{-\alpha/2} L_p$ with $\|g\|_{H_p^\alpha} = \|(1 - \Delta)^{\alpha/2} g\|_{L_p}$. B_ρ is the ball of radius ρ centered at the origin. For an integer $k \geq 0$, C_0^k denotes the C^k functions on R^d that vanish for $|x|$ sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_i u$ and $u_{x^i x^j} = D_{ij} u$. Repeated indices are summed. $L_p = L_p(R^d)$ is taken with respect to Lebesgue measure.

PDE

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noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) ->
F) :
  EuclideanSpace R (Fin d) -> F :=
  fun x => fderiv R u x (EuclideanSpace.single i 1)

noncomputable def multiDeriv (alpha : Fin d -> N) (u : EuclideanSpace R
(Fin d) -> F) :
  EuclideanSpace R (Fin d) -> F :=
  (List.finRange d).foldr (fun i v => (partialDeriv i)^[alpha i] v) u

noncomputable def sobolevNorm {d : N} (s : R) (p : R>=0inf) [Fact (1 <= p)]
(g : S'(EuclideanSpace R (Fin d), C)) : R>=0inf :=
  f in {f : Lp C p (volume : Measure (EuclideanSpace R (Fin d))) |
    bessePotential (EuclideanSpace R (Fin d)) C s g = f}, ||f||e

theorem krylov_sobolev_13_3_16_pointwise_decay_implies_negative_order_member
ship (d : N)
(hd : 0 < d) (p : R>=0inf) [Fact (1 <= p)] (hp1 : 1 < p) (hp2 : p !=
top) :
  (forall rho nu N0 : R, 0 < rho -> nu < d -> 0 < (nu + ) *
p.toReal -> (nu + ) * p.toReal < d ->
  exists C : R>=0inf, C != top /\ forall u : EuclideanSpace R (Fin d)
-> C, AESTronglyMeasurable u volume ->
  support u subset Metric.ball 0 rho -> (forall x != 0, ||u x|| <=
N0 * ||x|| ^ (-nu)) ->
  exists U : Lp C 1 (volume : Measure (EuclideanSpace R (Fin d))),
  u = ^m[volume] U /\
  MemSobolev p (U : S'(EuclideanSpace R (Fin d), C)) /\
  sobolevNorm p (U : S'(EuclideanSpace R (Fin d), C)) <= C) /\
forall (n : N) (rho nu N0 : R), 0 < rho -> nu < d -> <= n ->
  (( < n /\ 0 < (nu + - n) * p.toReal /\ (nu + - n) * p.toReal < d)
  /\
  ( = n /\ nu * p.toReal < d) ->
  exists C : R>=0inf, C != top /\ forall u : EuclideanSpace R (Fin d)
-> C, ContDiff R n u ->
  support u subset Metric.ball 0 rho ->
  (forall alpha : Fin d -> N, (sum i, alpha i) <= n -> forall x !=
0, ||multiDeriv alpha u x|| <= N0 * ||x|| ^ (-nu)) ->
  exists U : Lp C 1 (volume : Measure (EuclideanSpace R (Fin d))),
  u = ^m[volume] U /\
  MemSobolev p (U : S'(EuclideanSpace R (Fin d), C)) /\
  sobolevNorm p (U : S'(EuclideanSpace R (Fin d), C)) <= C :=
  by
sorry
```

Theorem 13.6.3. There exists a constant $\lambda_0 > 0$, depending only on d, p, κ, ω , and K , such that for any $\lambda \geq \lambda_0$ and $f^1, \dots, f^d, g$ in L_p there exists a unique u in W_p^1 satisfying (1). Furthermore, for this solution $\lambda^{1/2} \|u\|_{L_p} + \|Du\|_{L_p} \leq N(\lambda^{1/2} \|g\|_{L_p} + \sum_{i=1}^d \|f^i\|_{L_p})$, where N depends only on d, p, κ , and K . Notation: The standing setting of Section 13.6: p in $(1, \infty)$ is fixed and in R^d we consider the equation $Lu - \lambda u = D_i f^i + g$ where $Lu(x) = D_i(a^{ij}(x)D_j u(x) + a^i(x)u(x)) + b^i(x)D_i u(x) + c(x)u(x)$. All coefficients and f^i and g are real valued and, for some

PDE

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noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) ->
F) :
  EuclideanSpace R (Fin d) -> F :=
  fun x => fderiv R u x (EuclideanSpace.single i 1)

def HasWeakGradient {d : N} (u : EuclideanSpace R (Fin d) -> R)
(v : Fin d -> EuclideanSpace R (Fin d) -> R) : Prop :=
  forall (j : Fin d) (phi : EuclideanSpace R (Fin d) -> R), ContDiff R inf
phi -> HasCompactSupport phi ->
  (integral x, u x * partialDeriv j phi x) = -integral x, v j x * phi x
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Natural language statement

Domain

GroundTruth

constant $K, \kappa > 0$ and all i, j , on $\mathbb{R}^d$ $|a^{ij}|, |a^i|, |b^i|, |c| \leq K, a^{rk} x_i x_k \geq \kappa |x|^2$, for all x in $\mathbb{R}^d$. There also exists a function $\omega(\epsilon)$, $\epsilon > 0$, such that $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ and, for all i, j and x, y in $\mathbb{R}^d$ with $|x - y| \leq \epsilon$, $|a^{ij}(x) - a^{ij}(y)| \leq \omega(\epsilon)$. Solutions of (1) are sought in the class $W_{p,1} = H_{p,1}$, and (1) is understood in the sense of distributions. For an integer $k \geq 0$, C_0^k denotes the C^k functions on $\mathbb{R}^d$ that vanish for $|x|$ sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_{x^i} u$ and $u_{x^i x^j} = D_{x^i x^j} u$. Repeated indices are summed. $L_p = L_p(\mathbb{R}^d)$ is taken with respect to Lebesgue measure.

```
def IsDivergenceFormSolution {d : N} (a : Fin d -> Fin d -> EuclideanSpace
R (Fin d) -> R)
  (a' b : Fin d -> EuclideanSpace R (Fin d) -> R) (c : EuclideanSpace R
(Fin d) -> R) (lam : R)
  (f : Fin d -> EuclideanSpace R (Fin d) -> R) (g u : EuclideanSpace R
(Fin d) -> R)
  (v : Fin d -> EuclideanSpace R (Fin d) -> R) : Prop :=
HasWeakGradient u v /\
forall phi : EuclideanSpace R (Fin d) -> R, ContDiff R inf phi ->
HasCompactSupport phi ->
  ((sum i, integral x, ((sum j, a i j x * v j x) + a' i x * u x - f i
x) * partialDeriv i phi x) +
  integral x, g x * phi x) =
  integral x, ((sum i, b i x * v i x) + (c x - lam) * u x) * phi x

theorem krylov_sobolev_13_6_3_divergence_form_solvability {d : N} (hd : 0 <
d) (p : R) (hp1 : 1 < p) (hp2 : p != top) (kappa K : R) (hkappa :
0 < kappa) :
  exists N : R, forall w : R -> R, Tendsto w (nhds[>] (0 : R)) (nhds 0)
-> exists lam0 : R, 0 < lam0 /\
  forall (a : Fin d -> Fin d -> EuclideanSpace R (Fin d) -> R)
  (a' b : Fin d -> EuclideanSpace R (Fin d) -> R) (c : EuclideanSpace
R (Fin d) -> R),
  (forall i j, Measurable (a i j)) -> (forall i, Measurable (a' i))
-> (forall i, Measurable (b i)) ->
  Measurable c -> (forall i j x, |a i j x| <= K) -> (forall i x, |a'
i x| <= K) ->
  (forall i x, |b i x| <= K) -> (forall x, |c x| <= K) ->
  (forall x xi : EuclideanSpace R (Fin d), kappa * ||xi|| ^ 2 <= sum
r, sum k, a r k x * xi r * xi k) ->
  (forall eps : R, 0 < eps -> forall (i j : Fin d) (x y :
EuclideanSpace R (Fin d)), dist x y <= eps ->
  |a i j x - a i j y| <= w eps) ->
  forall lam : R, lam0 <= lam ->
  forall (f : Fin d -> Lp R p (volume : Measure (EuclideanSpace R
(Fin d))))
    (g : Lp R p (volume : Measure (EuclideanSpace R (Fin d))))
    (exists! u : Lp R p (volume : Measure (EuclideanSpace R (Fin d))))
    (exists v : Fin d -> Lp R p (volume : Measure (EuclideanSpace R
(Fin d))))
      IsDivergenceFormSolution a a' b c lam (fun i => (f i)) g u
fun j => (v j)) /\
  forall (u : Lp R p (volume : Measure (EuclideanSpace R (Fin d))))
    (v : Fin d -> Lp R p (volume : Measure (EuclideanSpace R (Fin
d))))
      IsDivergenceFormSolution a a' b c lam (fun i => (f i)) g u (fun
j => (v j)) ->
    Real.sqrt lam * ||u|| + (sum j, ||v j||) <=
      N * ((Real.sqrt lam)^(1 - p) * ||g|| + sum i, ||f i||) := by
sorry
```

Exercise 1.1.13. Let $m \geq 1$ be an integer and let a^α be some (complex) numbers, not all of which are zero, given for any multi-indices α such that $|\alpha| \leq m$. Consider the operator $L = \sum_{|\alpha| \leq m} a^\alpha D^\alpha$ and prove that the set $L C_0^\infty$ is everywhere dense in L_p for any p in $[2, \infty)$. Notation: A multi-index is a d -tuple $\alpha = (\alpha_1, \dots, \alpha_d)$ of non-negative integers, $|\alpha| = \alpha_1 + \dots + \alpha_d$, and $D^\alpha = D_1^{\alpha_1} \dots D_d^{\alpha_d}$. The exponent range here is $[2, \infty)$, not the $[1, \infty)$ of Theorem 1.1.6 Remark 1.1.14 flags the restriction. For an integer $k \geq 0$, C_0^k denotes the C^k functions on $\mathbb{R}^d$ that vanish for $|x|$ sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_{x^i} u$ and $u_{x^i x^j} = D_{x^i x^j} u$. Repeated indices are summed. $L_p = L_p(\mathbb{R}^d)$ is taken with respect to Lebesgue measure.

PDE

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noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) ->
F) :
  EuclideanSpace R (Fin d) -> F :=
fun x => fderiv R u x (EuclideanSpace.single i 1)

noncomputable def multiDeriv (alpha : Fin d -> N) (u : EuclideanSpace R
(Fin d) -> F) :
  EuclideanSpace R (Fin d) -> F :=
(List.finRange d).foldr (fun i v => (partialDeriv i)^[alpha i] v) u

theorem krylov_sobolev_1_13_const_coeff_operator_range_dense {d m : N}
(hm : 1 <= m)
  (P : MvPolynomial (Fin d) C) (hP : P != 0) (hPm : P.totalDegree <= m)
  (p : R) (hp : 2 <= p) :
  Dense {g : Lp C p (volume : Measure (EuclideanSpace R (Fin d))) |
exists phi : EuclideanSpace R (Fin d) -> C, ContDiff R (inf :
Ninfomega phi /\ HasCompactSupport phi /\
  (fun x => sum alpha in P.support, P.coeff alpha * multiDeriv
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Natural language statement

Domain

GroundTruth

		<pre>(alpha) phi x) =^m[volume] g} := by sorry</pre>
Exercise 1.3.23. (i) Let B be the open unit ball centered at the origin and let u be a twice continuously differentiable function on B-bar. Assume that $u = 0$ on ∂B. Set $f = \Delta u$ and prove that $\ u\ _{L^2(B)}^2 + \sum_i \ u_{x^i}\ _{L^2(B)}^2 \leq 4\ f\ _{L^2(B)}^2$. (ii) Given an integer n, denote by P_n the set of polynomials of x of degree $\leq n$ and let A be the operator $A : P_n \rightarrow P_n$ given by the formula $A_p = \Delta[(1 - x ^2)p]$. Conclude from (i) that A is invertible. Notation: Part (iii) of the exercise, which needs $W_2^2(B)$, is not formalized. $\Delta u = u_{x^1 x^1} + \dots + u_{x^d x^d}$ is the Laplacian, and "degree $\leq n$" means total degree in the d variables. For an integer $k \geq 0$, C_0^k denotes the C^k functions on $\mathbb{R}^d$ that vanish for x sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_{x^i} u$ and $u_{x^i x^j} = D_{ij} u$. Repeated indices are summed. $L_p = L_p(\mathbb{R}^d)$ is taken with respect to Lebesgue measure.	PDE	<pre>noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) -> F) : EuclideanSpace R (Fin d) -> F := fun x => fderiv R u x (EuclideanSpace.single i 1) theorem krylov_sobolev_1_3_23_unit_ball_estimate_and_polynomial_operator_bij ective {d : N} (hd : 1 <= d) : (forall u : EuclideanSpace R (Fin d) -> R, ContDiffOn R 2 u (Metric.closedBall 0 1) -> (forall x in Metric.sphere (0 : EuclideanSpace R (Fin d)) 1, u x = 0) -> eLpNorm u 2 (volume.restrict (Metric.ball 0 1)) ^ 2 + (sum i, eLpNorm (partialDeriv i u) 2 (volume.restrict (Metric.ball 0 1)) ^ 2) <= 4 * eLpNorm (Delta u) 2 (volume.restrict (Metric.ball 0 1)) ^ 2) /\ forall n : N, Set.BijOn (fun p : MvPolynomial (Fin d) R => sum i, MvPolynomial.pderiv i (MvPolynomial.pderiv i ((1 - (sum j, MvPolynomial.X j ^ 2)) * p))) {p : MvPolynomial (Fin d) R p.totalDegree <= n} {p : MvPolynomial (Fin d) R p.totalDegree <= n} := by sorry</pre>
Exercise 1.4.8. Let $d = 2$, $a^{ij}(x)$ be measurable functions on $\mathbb{R}^2$ satisfying $a^{ij} = a^{ji}$ and condition (5) for all x, x_i in $\mathbb{R}^2$, where $\mu > 0$ and $\nu > 0$ are some constants. For a $\lambda > 0$ define $L_\mu = L_\lambda$, $L_\nu = L_\lambda$, $a^{ij} u_{x^i x^j} - \lambda(a^{11} + a^{22})u$. Prove that, for any u in C_0^2, $\lambda(a^{11} + a^{22})\ u\ _{L^2}^2 + 2\lambda\mu\sum_{i,j=1}^2 \ u_{x^i x^j}\ _{L^2}^2 \leq (\nu/2)/(\mu/4)\ u\ _{L^2}^2$. Notation: Condition (5), from Exercise 1.4.7, is the two-sided ellipticity bound $\mu x ^2 \leq a^{ij}x_i x_j \leq \nu x ^2$ for all x in $\mathbb{R}^2$. For an integer $k \geq 0$, C_0^k denotes the C^k functions on $\mathbb{R}^d$ that vanish for x sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_{x^i} u$ and $u_{x^i x^j} = D_{ij} u$. Repeated indices are summed. $L_p = L_p(\mathbb{R}^d)$ is taken with respect to Lebesgue measure.	PDE	<pre>noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) -> F) : EuclideanSpace R (Fin d) -> F := fun x => fderiv R u x (EuclideanSpace.single i 1) theorem krylov_sobolev_1_4_8_two_dim_measurable_coeff_l2_estimate {mu nu lam : R} (hmu : 0 < mu) (hnu : 0 < nu) (hlam : 0 < lam) (a : Fin 2 -> Fin 2 -> EuclideanSpace R (Fin 2) -> R) (ha : forall i j, Measurable (a i j)) (hsymm : forall i j x, a i j x = a j i x) (hlb : forall (x : EuclideanSpace R (Fin 2)) (xi : Fin 2 -> R), mu * (sum i, xi i ^ 2) <= sum i, sum j, a i j x * xi i * xi j) (hub : forall (x : EuclideanSpace R (Fin 2)) (xi : Fin 2 -> R), (sum i, sum j, a i j x * xi i * xi j) <= nu * (sum i, xi i ^ 2)) (u : EuclideanSpace R (Fin 2) -> R) (hu : ContDiff R 2 u) (hu0 : HasCompactSupport u) : ENNReal.ofReal (lam ^ 2) * eLpNorm u 2 volume ^ 2 + ENNReal.ofReal (2 * lam) * (sum j, eLpNorm (partialDeriv j u) 2 volume ^ 2) + (sum j, sum k, eLpNorm (partialDeriv k (partialDeriv j u)) 2 volume ^ 2) <= ENNReal.ofReal (nu ^ 2 / mu ^ 4) * eLpNorm (fun x => (sum i, sum j, a i j x * partialDeriv j (partialDeriv i u) x) - lam * (a 0 0 x + a 1 1 x) * u x) 2 volume ^ 2 := by sorry</pre>
Exercise 9.1.7. Prove that, if u in L_2 and $\int_0^1 \ u^{(k)}\ _{L_2}^2 dx \leq M^2$, then u in W_2^1 and $\ u_x\ _{L_2} \leq N(M + \ u\ _{L_2})$, where N is independent of M and u. Notation: The mollification (1.8.4) is $u^{(k)}(x) = \int_{\mathbb{R}^d} u(x - \epsilon y) \zeta_k(y) dy$, and the standing assumption inherited from Exercise 9.1.6 is that ζ_k in C_0^∞ is even and integrates to one. W_2^1 is the space of u in L_2 whose generalized first derivatives u_{x^i} exist and lie in L_2. v is the generalized derivative $D_{x^i} u$ when $\int u D_{x^i} \phi = -\int v \phi$ for every ϕ in C_0^∞. $\ u_x\ _{L_2}$ is the summed first-order seminorm $\sum_i \ u_{x^i}\ _{L_2}$. For an integer $k \geq 0$, C_0^k denotes the C^k functions on $\mathbb{R}^d$ that vanish for x sufficiently large, and C_0^∞ the infinitely differentiable ones. Subscripts denote partial derivatives: $u_{x^i} = D_{x^i} u$ and $u_{x^i x^j} = D_{ij} u$. Repeated indices are summed. $L_p = L_p(\mathbb{R}^d)$ is taken with respect to Lebesgue	PDE	<pre>noncomputable def partialDeriv (i : Fin d) (u : EuclideanSpace R (Fin d) -> F) : EuclideanSpace R (Fin d) -> F := fun x => fderiv R u x (EuclideanSpace.single i 1) def HasWeakGradient {d : N} (u : EuclideanSpace R (Fin d) -> R) (v : Fin d -> EuclideanSpace R (Fin d) -> R) : Prop := forall (j : Fin d) (phi : EuclideanSpace R (Fin d) -> R), ContDiff R inf phi -> HasCompactSupport phi -> (integral x, u x * partialDeriv j phi x) = -integral x, v j x * phi x theorem krylov_sobolev_9_1_7_mollification_rate_implies_weak_derivative {d : N} (zeta : EuclideanSpace R (Fin d) -> R) (hzetaSmooth : ContDiff R inf zeta) (hzetaSupp : HasCompactSupport zeta) (hzetaEven : forall y, zeta (-y) =</pre>

Natural language statement

Domain

GroundTruth

measure.		<pre>zeta y) (hzetaone : (integral y, zeta y) = 1) : exists N : R>=0, forall (u : EuclideanSpace R (Fin d) -> R) (M : R>=0), MemLp u 2 volume -> (integral^~ eps in Ioo (0 : R) 1, eLpNorm (fun x => (integral y, u (x - eps * y) * zeta y) - u x) 2 volume ^ 2 * ENNReal.ofReal (eps ^ (-3 : Z))) <= (M : R>=0inf) ^ 2 -> exists v : Fin d -> EuclideanSpace R (Fin d) -> R, HasWeakGradient u v /\ (forall j, MemLp (v j) 2 volume) /\ sum j, eLpNorm (v j) 2 volume <= (N : R>=0inf) * ((M : R>=0inf) + eLpNorm u 2 volume) := by sorry</pre>
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